FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – VIII
PAPER –2
TEST DATE: 07-05-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B

Sol. Flux enclosed is

$$\phi = \frac{q_{in}}{\varepsilon_0} = \frac{\rho\left(\frac{1}{2}\cdot 8R\cdot 3R\right)R}{\varepsilon_0}$$

$$\phi = \frac{12\rho R^3}{\varepsilon_0}$$

2. C

Sol. Force weight of liquid above the sphere

$$F = P_0\pi r^2 + (\pi r^2\cdot 3r + \pi r^3 - \frac{2}{3}\pi r^3)\rho g$$

$$= P_0\pi r^2 + \frac{10}{3}\pi r^3\rho g = \frac{\pi r^2}{3}(3P_0 + 10\rho g r)$$

3. B

Sol. Since current is

$$I = \vec{J}\cdot\vec{S} = JS = \sigma ES = KE^2\cdot 4\pi r^2$$

$$\therefore\ E = \frac{1}{r}\sqrt{\frac{I}{4\pi K}}$$

$$\text{Also, } V = -\int_b^a \vec{E}\cdot d\vec{r} = \sqrt{\frac{I}{4\pi K}}\ell n\left(\frac{b}{a}\right)$$

$$\therefore\ I = \frac{4\pi KV^2}{\left\{\ell n\left(\frac{b}{a}\right)\right\}^2}$$

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4. C

Sol. Let density be ρ, where $\rho = \dfrac{2M}{\frac{4}{3}\pi\left[(2R)^3 - (R)^3\right]}$

$$\rho = \frac{2M}{\frac{4}{3}\pi(7R^3)}$$

Moment of inertia of remaining part (about point of contact)

$I = I_T - I_{cavity}$

$$I = \frac{7}{5}M_T(2R)^2 - \frac{7}{5}M_CR^2$$

$$M_T = \frac{16M}{7} \text{ and } M_C = \frac{2M}{7}$$

$$I = \frac{62}{5}MR^2$$

$$\text{K.E.} = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{62}{5}\right)MR^2\left(\frac{V}{2R}\right)^2$$

$$\text{K.E.} = \frac{31}{20}Mv^2$$

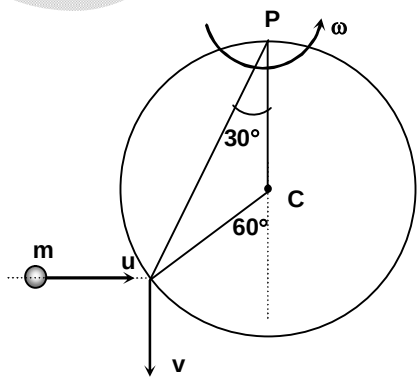
5. B, D

Sol. usin 30° = v sin60°

$$\Rightarrow v = \frac{u}{\sqrt{3}} \quad \ldots(i)$$

$$mu \times \frac{3R}{2} = mv \times \frac{\sqrt{3}R}{2} + \frac{3}{2}mR^2\omega$$

$$\Rightarrow \omega = \frac{2u}{3R} \quad \ldots(ii)$$



Coefficient of restitution,

$$e = \frac{\frac{v}{2} + \frac{\sqrt{3}}{2}\omega R}{\frac{\sqrt{3}}{2}u} = \frac{v + \sqrt{3}\omega R}{\sqrt{3}u}$$

Or e = 1

6. A, C

Sol. zero error = – 1.25 mm

Reading = 18 + 0.34 + 1.25 = 19.59 mm

7. C, D

Sol. At P, path difference

$$\Delta x = \left(\frac{4}{3} - 1\right)9\lambda - \left(\frac{3}{2} - 1\right)2\lambda = 2\lambda$$

$I_P = 4I_0$

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8. B, C, D

Sol. $\ell n\left(\frac{v_n}{v_1}\right) = -3\ell nn$

$\ell n\left(\frac{r_n}{r_1}\right) = 2\ell nn$

$\ell_h\left(\frac{A_n}{A_1}\right) = 4\ell nn$

$r_h = r_0\left(\frac{n^2}{z}\right)$

9. A, D

Sol. $\frac{qE\ell}{2} = \frac{1}{2}mv^2$

$\therefore\ v = \sqrt{\frac{qE\ell}{m}}$

$a = \frac{v^2}{\ell} = \frac{qE}{m}$

$T - qE = \frac{m}{\ell}\left(\frac{qE\ell}{m}\right)$

$\therefore\ T = 2qE$

10. B, C

Sol. For A,

$J - J_1 = mu$ …(i)

For C,

$J_2 = mv$ …(ii)

Component of velocity of B in direction BA must be u.

Let its vertical component perpendicular to BA be v_1

Velocity component of B along CB is

$u\sin 30° - v_1\cos 30° = v$ …(ii)

For B

$J_1 - J_2\cos 60° = 2mu$

$2J_1 - J_2 = 4mu$ …(iv)

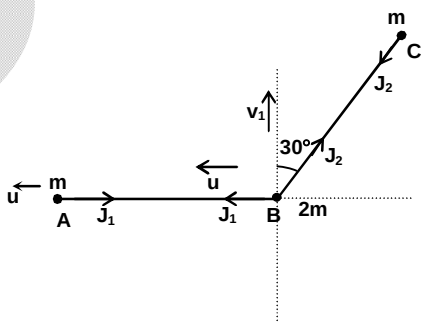
and $J_2\cos 30° = 2mv_1$

$\frac{\sqrt{3}}{2}J_2 = 2mv_1$ …(v)

Solving (ii), (iii) and (iv) and (v)

$v = \frac{4u}{11}$, $v_1 = \frac{\sqrt{3}u}{11}$ and $v_B = \sqrt{u^2 + v_1^2}$

$v_B = \frac{4\sqrt{31}\,u}{11}$



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Section – B

11. 5

Sol. $F_A = \frac{2m}{3}\omega^2 \times \frac{2\ell}{3}$, $F_C = \frac{m}{2}\omega^2 \times \frac{3\ell}{4}$

$\frac{F_A}{F_C} = \frac{32}{27} \Rightarrow n = 5$

12. 2

Sol. Elastic potential energy,

$U = \frac{1}{2}\left(\frac{YA}{\ell}\right)(\Delta\ell)^2$

Here y, ℓ, $\Delta\ell$ are same for both the wire

$\Rightarrow \frac{U_B}{U_A} = \frac{S_B}{S_A} = 2$

13. 2

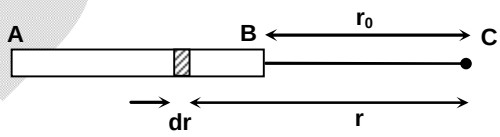
Sol. $I_0 = \frac{k_0 A}{\cos\left(\frac{\pi}{6L}x\right)}\frac{dT}{dx}$

$\Rightarrow T = T_0 \frac{6I_0 L}{\pi k_0 A}\sin\left(\frac{\pi}{6L}x\right)$

$\Rightarrow T = 2T_0$ at $x = L$

14. 5

Sol. Potential due to rod at C



$V = -\frac{GM}{L}\int_{r_0}^{r_0+L}\frac{dx}{x}$

$v = -\frac{GM}{L}\ell n\left(1+\frac{L}{r_0}\right)$, where r_0 changes from L to $\frac{L}{2}$ then kinetic energy gained by m is

$\frac{1}{2}mu^2 = \frac{mGM}{L}\ell n\left(\frac{3}{2}\right)$

$v = \sqrt{\frac{2GM}{L}\ell n\left(\frac{3}{2}\right)}$

$\therefore x + y = 2 + 3 = 5$

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15. 4

Sol. $2T\sin\theta = BidR$

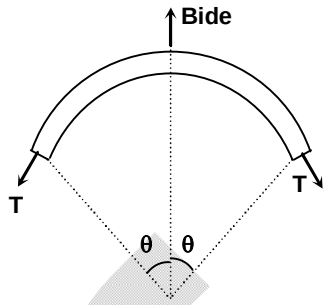
$2T\theta = Bi(R \times 2\theta)$

$T = BiR$

$$v = \sqrt{\frac{T}{\rho S}} = \sqrt{\frac{BiR}{\rho S}}$$

$$B = \frac{V^2\rho S}{iR} = \frac{10\times10\times2\times10^3\times0.2\times10^{-4}}{1\times1}$$

$B = 4T$



16. 2

Sol. Amplitude of block = $\frac{mg}{k} = 1\,mm$

$$\frac{1}{v} + \frac{1}{(-30)} = \frac{-1}{20} \Rightarrow v = -60\ cm$$

Magnification, $m = \frac{-v}{u} = -2$

Thus, amplitude of image = 2 mm

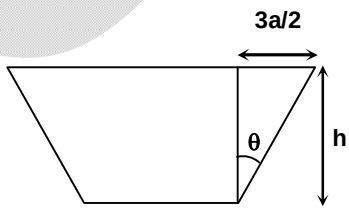
17. 3

Sol. $\tan\theta = \frac{3a}{2h}$

$2 \times 4aT \times \cos\theta = 4a\lambda g$

$$\Rightarrow \cos\theta = \frac{\lambda g}{2T}$$

$$\Rightarrow h = \frac{3\lambda g a}{2\sqrt{4T^2 - \lambda^2 g^2}}$$



18. 4

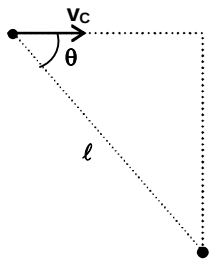
Sol. $$\frac{v}{(v - v_C\cos\theta)}(1000) = 1500$$

$\Rightarrow \theta = 30$

Also, $$\frac{\ell}{v} + \frac{\ell\sin\theta}{v_{bullet}} = \frac{\ell\cos\theta}{v_C}$$

$$\Rightarrow \frac{1}{2v_{bullet}} = \frac{1}{v}\left(\frac{\sqrt{3}v}{2v_C} - 1\right) = \frac{5}{4v}$$

$$\Rightarrow v_{bullet} = \frac{2v}{5} = \frac{2}{5}(360) = 4(36)\ m/s$$



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Chemistry

PART – II

Section – A

19. D

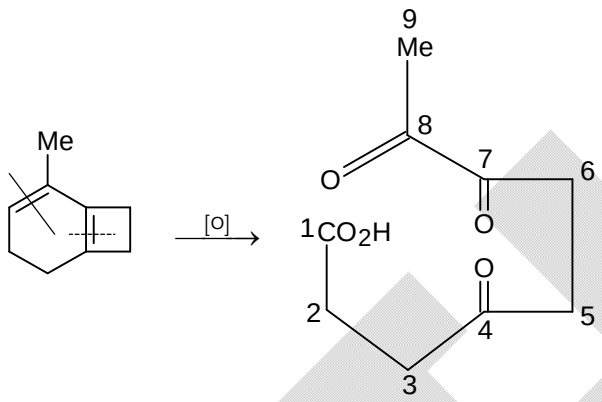
Sol. $K_P = \frac{n_{PCl_3} \times n_{Cl_2}}{n_{PCl_5}} \left(\frac{P}{\sum n}\right)^{\Delta n}$

$= \frac{1\times 1}{4} \times \left(\frac{1}{7}\right)^1$

$= \frac{1}{28}$

20. C

Sol.



4, 7, 8-trioxo nonanoic acid

21. B

Sol. In polar protic solvent down the group nucleophilicity increases and when the attacking atom is same generally stronger the base stronger the nucleophile provided there is no unfavourable steric hindrance.

22. B

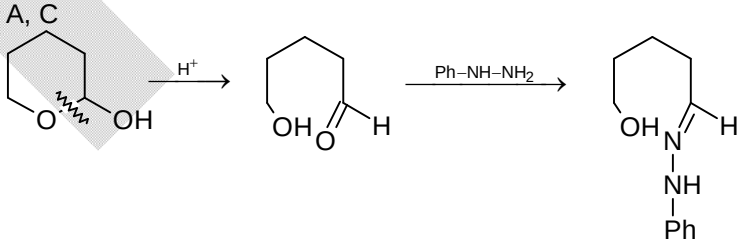
Sol. Alanine is a neutral amino acid. The isoelectric points of neutral amino acids are in the pH range 5.5 to 6.3

23. B, D

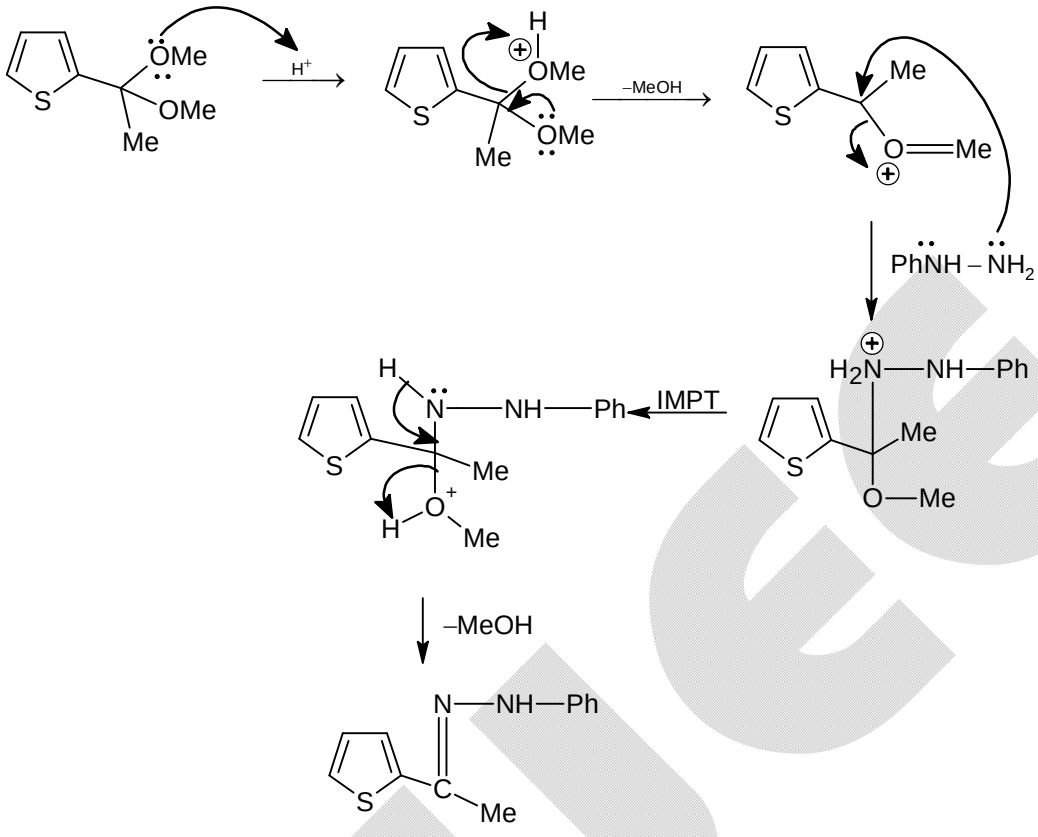
Sol. If the diol/polyol involves the OH group oriented in such a way that they accurately match the structural parameters required by a tetrahedrally coordinated boron, a strong complex is formed.

24. A, C

Sol.

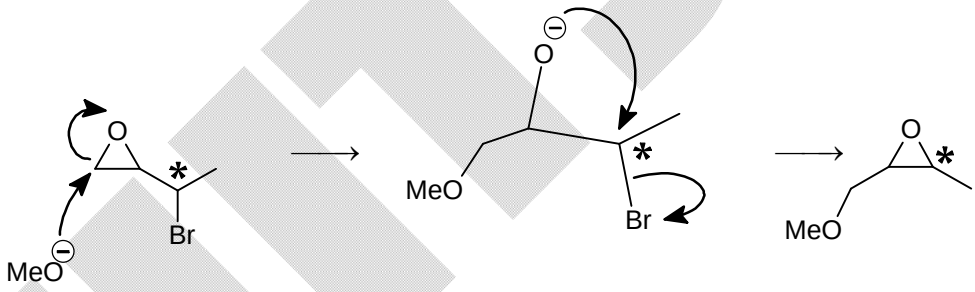


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25. C, D

Sol.



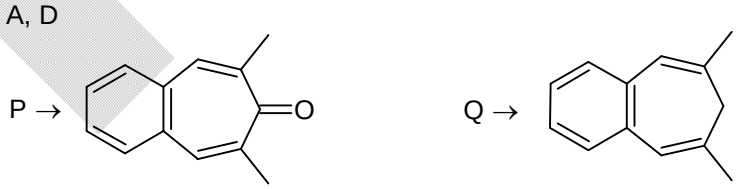
26. B, D

Sol. $b_{N_2} > b_{O_2}$

Since volume of N_2 molecule is greater than O_2 molecule. Both are non-polar molecule, in that case a is proportional to molecular mass.

27. A, D

Sol.



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28. A, C, D

Sol. (A) $\Delta T_f = K_f \times m$

$= 1.86 \times \frac{0.04}{200} \times 1000$

$= 0.372$

(B) $\Delta G^- = -RT \ln K_{eq}$

$\Delta G^- = \Delta H^- - T\Delta S^-$

$\ln K_{eq} = \frac{\Delta H^-}{RT} + \frac{\Delta S^o}{R}$

(C) $A(g) \longrightarrow 2B(g) + C(g)$

$t = 0$	a		
$t = t_{eq}$	$a - x$	$2x$	x

$P_o \propto a$

$P_t \propto a + 2x$

$x \propto \frac{P_t - P_o}{2}$

$a - x \propto P_o - \left(\frac{\{P_t - P_o}{2}\right)$

$\propto \frac{3P_o - P_t}{2}$

$k = \frac{1}{t} \ell n \frac{a}{a - x}$

$k = \frac{1}{t} \ell n \frac{P_o}{\frac{3P_o - P_t}{2}}$

$k = \frac{1}{t} \ell n \frac{2P_o}{3P_o - P_t}$

(D) $2\pi r = n\lambda$

$2\pi \times 0.5n^2 = n \times 12.56\,\text{Å}$

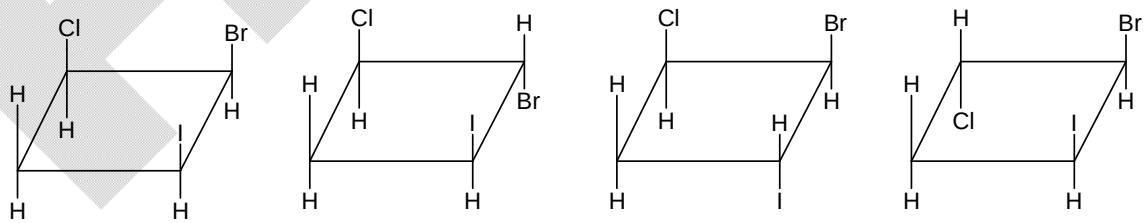
$3.14 \times n = 12.56$

$n = 4$

Section – B

29. 1

Sol.



P has four diastereomers and each of the diastereomers exhibits enantiomerism.

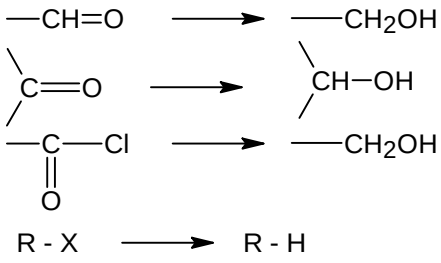
30. 5

Sol. CO_2, CH_3CN, NO_2^+, N_2O & C_3O_2

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31. 7

Sol. $NaBH_4$ can reduce only following group.



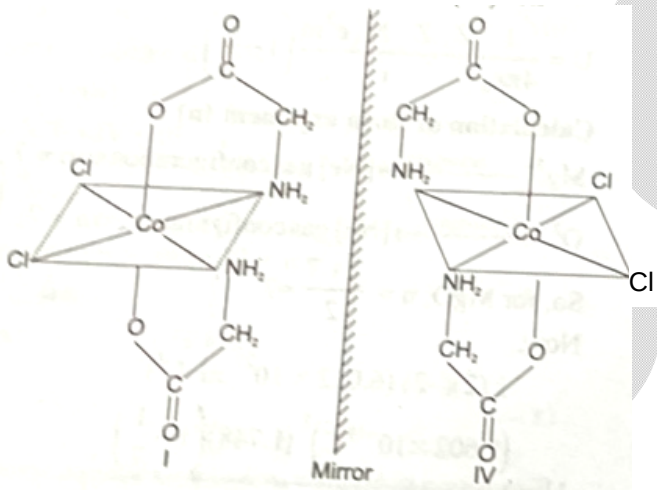
R - X ⟶ R - H

($2^o/3^o$ halide)

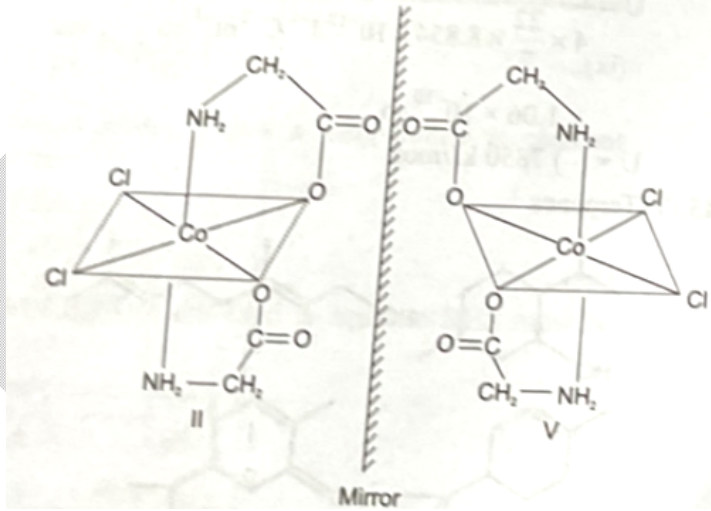
32. 6

Sol.

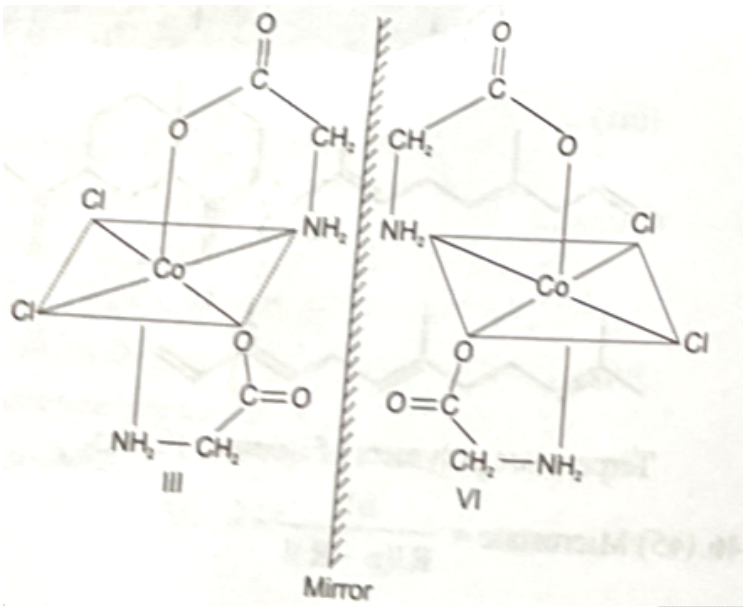
Dichloridobis (glycinato) cobaltate (III) ion $\left[CoCl_2(gly)_2\right]^{\ominus}$



Dichloridobis (glycinato) cobaltate (III) ion



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Total number of optical active isomers = 6(six)

33. 5

Sol.

$$\begin{array}{lll} & 7A \longrightarrow & B \\ t=0 & a_0 & \\ t=t & a_0-7x & x \end{array}$$

Now, $a_0 = 8x$

$$8x \xrightarrow{t_{1/2}} 4x \xrightarrow{t_{1/2}} 2x \xrightarrow{t_{1/2}} x$$

Given $3t_{1/2} = 900$ hr

$t_{1/2} = 300$ hr

$$= \frac{300}{24} \text{ days}$$

$$\therefore 0.4t_{1/2} = \frac{300}{24} \times \frac{4}{10} \text{ days}$$

= 5 days

34. 4

Sol. In the given molecule there are 4 acidic hydrogen atoms are present.

35. 9

Sol. $FeSO_4.(NH_4)_2SO_4.6H_2O \longrightarrow Fe^{2+} + 2NH_4^+ + 2SO_4^{2-}$

x = 5

$$2CuSO_4 + 4KCN \longrightarrow 2CuCN + K_2SO_4 + (CN)_2 \uparrow$$

$$3KCN + CuCN \longrightarrow K_3[Cu(CN)_4]$$

$$K_3[Cu(CN)_4] \longrightarrow +3K^+ + [Cu(CN)_4]^{3-}$$

y = 4

36. 4

Sol. Lattice energy $\propto$ charge of cation × charge of anion.

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Mathematics

PART – III

Section – A

37. C

Sol. $T_n = 2 + \frac{1}{n} - \frac{1}{n+2}$

38. D

Sol. Use the function $f(x) = (\cot^{-1}(x))^2 + \frac{2}{\sqrt{x^2+1}}$

39. A

Sol. Sum of elements = $\sum_{r=0}^{3}\left(\frac{4\pi}{3} + 2r\pi\right) = \frac{52\pi}{3}$

40. C

Sol. Put $e^{\frac{y^2}{x}} = t \Rightarrow y(y-x) = x\log_e(ce^y - 1)$

41. A, D

Sol. $A^4 - 7A^3 + 15A^2 - 9A = 0$ (Null matrix)

$\Rightarrow B + C = 4A$

42. A, B, C, D

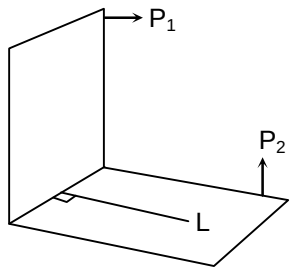
Sol. $P_1 : 8x - 9y + 4z = -40$

$P_2 : 3x + 4y + 3z = -10$

Line L must lie in P_2 and perpendicular to P_1

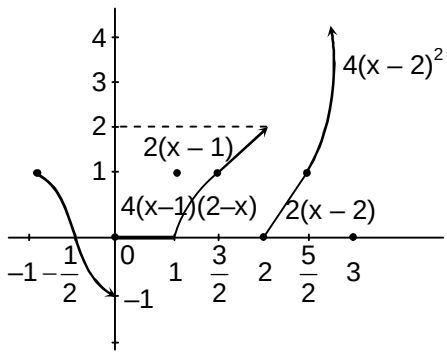
$\therefore A\left(x_1, y_1, \frac{10}{3}\right)$ must lie in both the plane

$\therefore x_1 = -\frac{20}{3}, y_1 = 0, z_1 = \frac{10}{3}$



43. A, B, C

Sol. Shown in the figure



44. B, C

Sol. $f(\alpha) = \frac{\alpha}{\sqrt{1-\alpha^2}}$

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45. A, B, C, D

Sol. $\lim\limits_{h\to0}\dfrac{(1+h)^a-5(1+h)+4}{h\left\{(1+h)^a-4(1+h)+3\right\}}=\lim\limits_{h\to0}\dfrac{\left[\left\{1+ah+a\left(\dfrac{a-1}{2}\right)h^2+.....\right\}-5(1+h)+4\right]}{h\left[\left\{(1+ah)+a\left(\dfrac{a-1}{2}\right)h^2+.....\right\}-4(1+h)+3\right]}$

$=\lim\limits_{h\to0}\dfrac{(a-5)h+a\left(\dfrac{a-1}{2}\right)h^2+.....}{h\left\{(a-4)h+a\left(\dfrac{a-1}{2}\right)h^2+.....\right\}}\Rightarrow(a=5)\text{ and }b=\dfrac{a\left(\dfrac{a-1}{2}\right)}{a-4}=10$

46. A, D

Sol. Let $z_1=2e^{ia},z_2=\dfrac{14\sqrt2}{5}e^{ic},z_3=2e^{ib}$

$\therefore\ \arg(z_2-z_1)=\alpha\Rightarrow\dfrac{\frac{14\sqrt2}{5}\sin(a+\theta)-2\sin a}{\frac{14\sqrt2}{5}\cos(a+\theta)-2\cos a}=\dfrac{4}{3}\ ;\ \cos\theta=\dfrac{7\sqrt2}{10}$

$\Rightarrow\tan a=\dfrac{3}{4}\ \therefore\ \tan(a+2\theta)=\dfrac{4}{3}\ \therefore\ \arg(z_1)+\arg(z_3)=\dfrac{\pi}{2}$

Section – B

47. 4

Sol. Number of favourable cases = 7788

48. 5

Sol. $I=\int_0^1\dfrac{x^{15}-x^{11}+x^7}{\left(3x^{16}-4x^{12}+6x^8\right)^{\frac34}}dx\Rightarrow I=\dfrac{1}{48}\int_0^5\dfrac{dt}{t^{\frac34}}=\dfrac{5^{\frac14}}{12}$

49. 2

Sol. Let $\cot^{-1}(2x-1)=\theta\ ;\ 0<\theta<\pi$

$\Rightarrow\sin2\theta=\dfrac{1+\cot\theta}{2}\Rightarrow\dfrac{2t}{1+t^2}=\dfrac{t+1}{2t}\Rightarrow t^3-3t^2+t+1=0$

$\Rightarrow t=1,1\pm\sqrt2\ ;\ t=1-\sqrt2$ does not satisfy

$\therefore\ x=1,\dfrac{1}{\sqrt2}$

50. 6

Sol. $\dfrac{1}{{}^{2024}C_r}=\dfrac{2025}{2026}\left\{\dfrac{1}{{}^{2025}C_r}+\dfrac{1}{{}^{2025}C_{r+1}}\right\}$

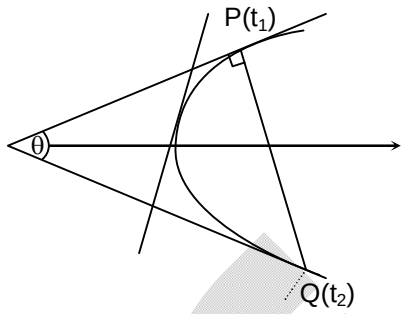
$\Rightarrow\sum_{r=1}^{2023}\dfrac{(-1)^{r-1}r}{{}^{2024}C_r}=\dfrac{2025}{2026}\left\{\left(\dfrac{1}{C_1}+\dfrac{1}{C_2}\right)-2\left(\dfrac{1}{C_2}+\dfrac{1}{C_3}\right)+3\left(\dfrac{1}{C_3}+\dfrac{1}{C_4}\right)+.....2023\left(\dfrac{1}{C_{2023}}+\dfrac{1}{C_{2024}}\right)\right\}$

$=\dfrac{2025}{2026}\left\{\dfrac{1}{C_1}-\dfrac{1}{C_1}+\dfrac{1}{C_3}-\dfrac{1}{C_4}+....-\dfrac{1}{C_{2024}}+\dfrac{2024}{C_{2024}}\right\}=\dfrac{2024}{2026}=\dfrac{1012}{1013}$

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51. 2

Sol. $PQ = \dfrac{4a\left(t_1^2+1\right)^{\frac{3}{2}}}{t_1^2}$



PQ is max at $t_1 = 3$ and $t_2 = -3-\dfrac{2}{3} = -\dfrac{11}{3}$

$$\tan\theta = \left|\frac{\frac{1}{3}+\frac{3}{11}}{1-\frac{1}{3}\cdot\frac{3}{11}}\right| = \frac{20}{33}\times\frac{11}{10} = \frac{2}{3}$$

52. 6

Sol. $\vec{r} = x\vec{a} + y\vec{b} + z\left(\vec{a}\times\vec{b}\right)$ ..... (1)

$\vec{r}\times\left(\vec{a}\times\vec{b}\right) = \left(\vec{r}\cdot\vec{b}\right)\vec{a} - \left(\vec{r}-\vec{a}\right)\vec{b} \Rightarrow \vec{r}\cdot\vec{b} = 3,\ \vec{r}\cdot\vec{a} = 2$

From equation (1), we get $2 = x|a|^2 + y\vec{a}\cdot\vec{b} \Rightarrow 2x + y = 1$

$3 = x\vec{a}\cdot\vec{b} + y|b|^2 \Rightarrow 2x + 4y = 3$

$\therefore\ x = \dfrac{1}{6}; y = \dfrac{2}{3}$. Also, $|\vec{r}| = \dfrac{4}{\sqrt{3}} \Rightarrow z = \pm\dfrac{1}{2}$

$\therefore\ \left|\left[\vec{r}\ \ \vec{a}\ \ \vec{b}\right]\right| = |z||a\times b|^2$

53. 7

Sol. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$; $ad - bc \neq 0 \Rightarrow a + b + c + d = 6$

(A) Exactly one zero. Number of matrices = $\dfrac{4!}{2!} + 4! + \dfrac{4!}{3!}$

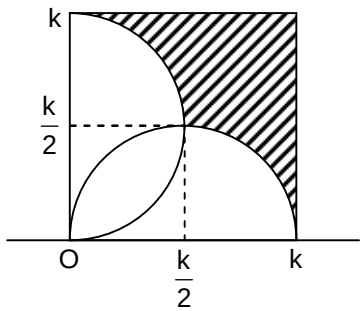
(B) Exactly two zeros. Number of matrices $2\cdot{}^{6-1}C_{2-1} = 10$

(C) No zeroes number of matrices = 6

54. 6

Sol. Let z be x + iy then $0 < x < k$, $0 < y < k$

And $x^2 + y^2 > kx$, $x^2 + y^2 > ky$



$$\text{Area} = k^2 - \left[\frac{\pi k^2}{4} - 2\left\{\frac{\pi}{4}\left(\frac{k}{2}\right)^2 - \frac{1}{2}\cdot\frac{k^2}{4}\right\}\right]$$

$$= \frac{3}{4}k^2 - \frac{\pi}{2}\cdot\frac{k^2}{4} = \frac{k^2}{4}\left(3-\frac{\pi}{2}\right) = 250(6-\pi)$$

$\Rightarrow k^2 = 2000 \Rightarrow k = 20\sqrt{5}$

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